Repeat Purchase Prediction

Pipeline Structure and Rationale

Group 22 • Machine Learning, NOVA IMS
dataset “Everything & Then Some”

Handout PDF

Members
Mohamed Triki • Wael Farhat • Gustavo Carvalho

Kaggle team ML_Group_22

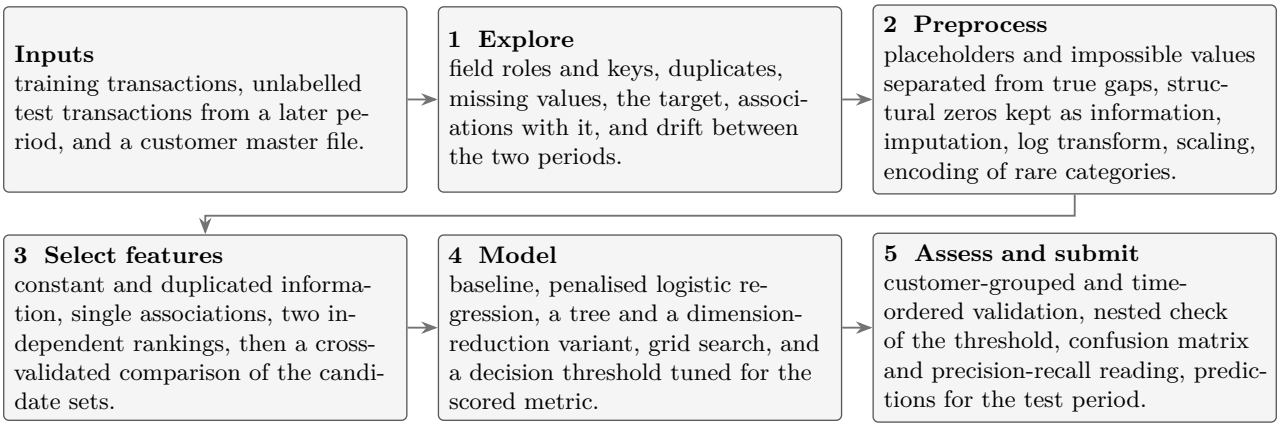
1. Objective and data

The task is to predict whether the customer behind a purchase will buy again within ninety days. Each row is one transaction, the answer is yes or no, and the competition scores the positive class with F1, so the work is judged on how reliably repeaters are found, not on how often the model is right overall.

Training covers 6,534 transactions from 610 customers between July 2015 and February 2025; the test file covers 986 transactions from 206 customers between May 2025 and October 2026 and carries no label. A customer file describes 1,143 accounts and joins on the customer key. The two transaction files share customers but not rows, and 94.5% of test rows belong to a customer the company has already seen, which is the most important fact about the evaluation. After cleaning, 56% of transactions repeat, so the classes are close in size and always answering yes already scores 0.718.

2. Structure of the pipeline

Four notebooks, each handing its output to the next as a plain table.



3. The stages and why each one suits this data

Preprocessing. A placeholder for an unknown amount is not a large purchase, so those 88 cells and the impossible negatives became missing values, flagged rather than clipped. An empty purchase history is not missing at all: 1,048 rows belong to customers with no earlier order, so those fields were set to zero and marked. Amounts are skewed, so a log transform and scaling follow, and rare categories were grouped before encoding.

Integrating the customer data. The customer file is optional, so it was joined only where it earns its place, and the key was checked first so no transaction is duplicated. Region and cohort carry real signal: one region repeats on 94.5% of its rows, almost 39 points above the average. Street, company name and postal code were left out: identifiers dressed as features, which would let the model memorise addresses instead of learning behaviour.

Feature selection. Selection was made by concept rather than by column, because encoding one category as several indicators makes those columns dependent. Constant and near-duplicate fields go first, then weak single associations, then two rankings that disagree in useful ways, and finally a cross-validated comparison that prefers the smaller set. The funnel runs from 53 concepts and 124 columns to 19 and 64.

Modelling. With a few thousand rows of tabular, business-readable fields, a penalised logistic regression is the honest first choice: stable at this size, lightly tuned, and one interpretable coefficient per feature. A decision tree and a dimension-reduction variant were fitted as checks beside an always-yes baseline; the linear model won, 0.748 against 0.723 and 0.745.

Assessment. The same customers appear many times, so a random split tests the model on rows whose customer already helped fit it; the folds are grouped on the customer. Because the score depends on a decision threshold as much as on the model, that threshold was tuned on out-of-fold probabilities and re-verified inside a nested loop. A final check fits the earlier years and scores the later ones.

4. Results

The grouped 0.755 is the honest floor, since those folds use customers the model has never met; the time-ordered 0.795 is the realistic expectation, and the submission's 0.782 sits between them.

Measure	F1
Always-yes baseline	0.718
Customer-grouped cross-validation	0.755
Time-ordered check, all rows	0.795
returning customers	0.801
first appearance	0.730
Kaggle, public part of the test set	0.782

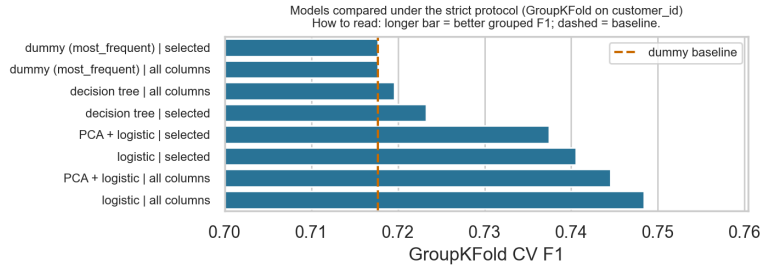


Figure 1. Every model family under the strict customer-grouped protocol, against the baseline. The bars sit close together: at this size the honesty of the split matters more than the choice of algorithm.

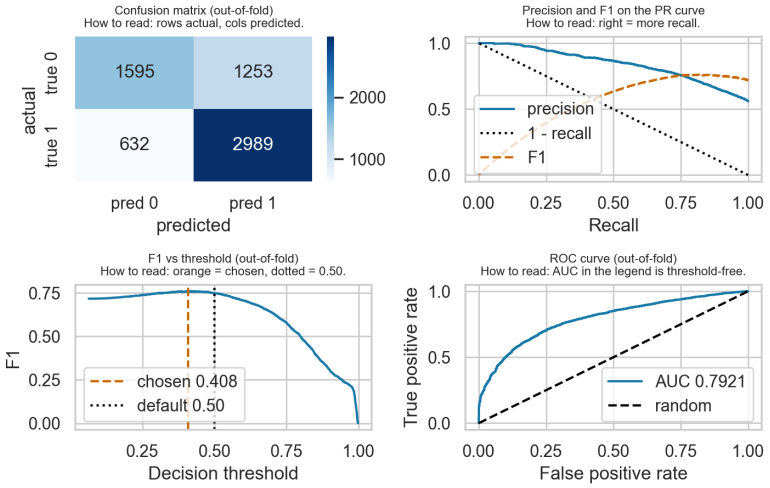


Figure 2. The assessment read together: the confusion matrix at the chosen threshold, the precision-recall trade, the score against the threshold, and the ranking of probabilities on its own.

Annexes

A. Reproducing the work. The four notebooks run in sequence in about a minute with a fixed random seed, and every stage writes its result as a readable table, so any number can be checked afterwards. Only tools covered in the course were used: no resampling, no bagging, no boosting.

B. Choices that were tested and then dropped. Customer-level history features, ratios, winsorising, quartile binning and interactions were all tried; none beat the noise between folds, and the customer aggregates made things worse. Scaling the yes-no flags as well as the amounts was also worse.

C. Limitations. The threshold was chosen on the training period and predicts more repeaters than that period holds, so it is the first value to revisit if the private leaderboard disagrees.